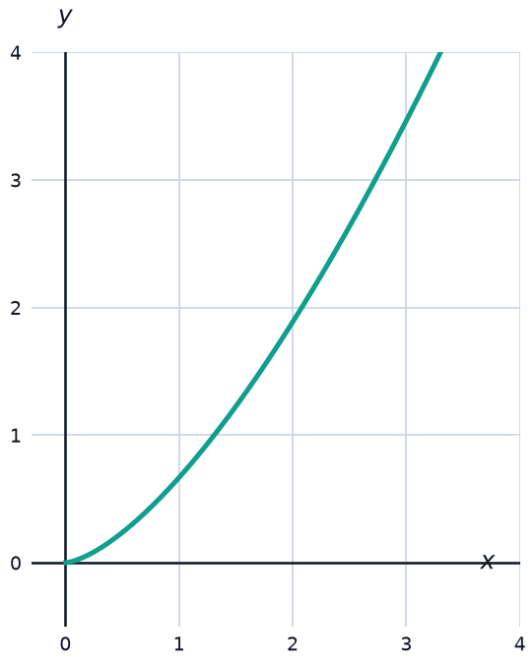
Arc Length and Improper Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

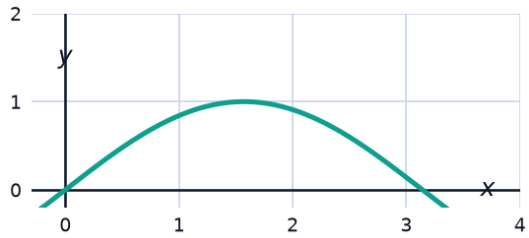
1 [Foundational] • Target: Arc length

Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.



2 [Foundational] • Target: Arc length

Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.



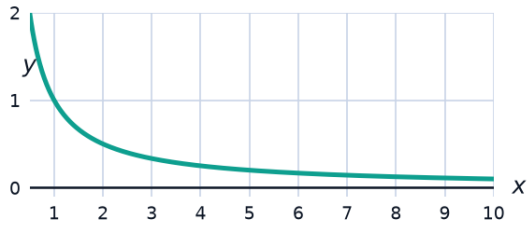
3 [Foundational] • Target: Improper integrals

Evaluate $\int_1^{\infty} \frac{1}{x^2}\,dx$, or state that it diverges.

© ClipSAT — clipsat.org | All Rights Reserved

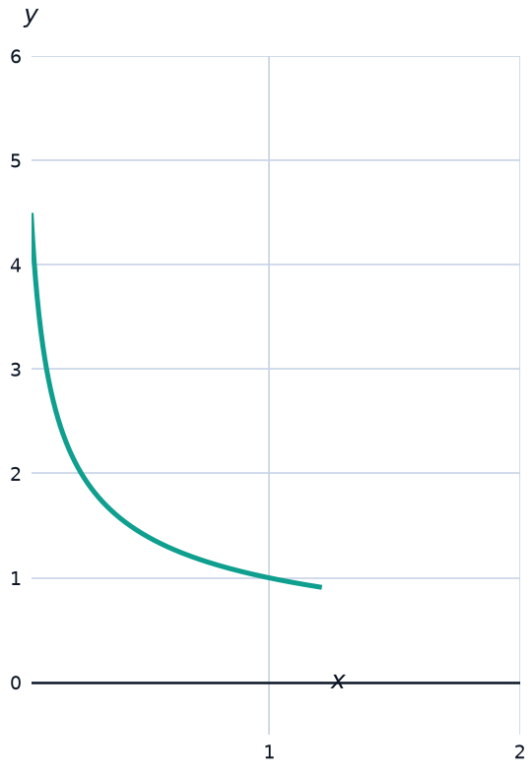
4 [Foundational] • Target: Improper integrals

Evaluate $\int_1^{\infty} \frac{1}{x}\,dx$, or state that it diverges.



5 [Foundational] • Target: Improper integrals

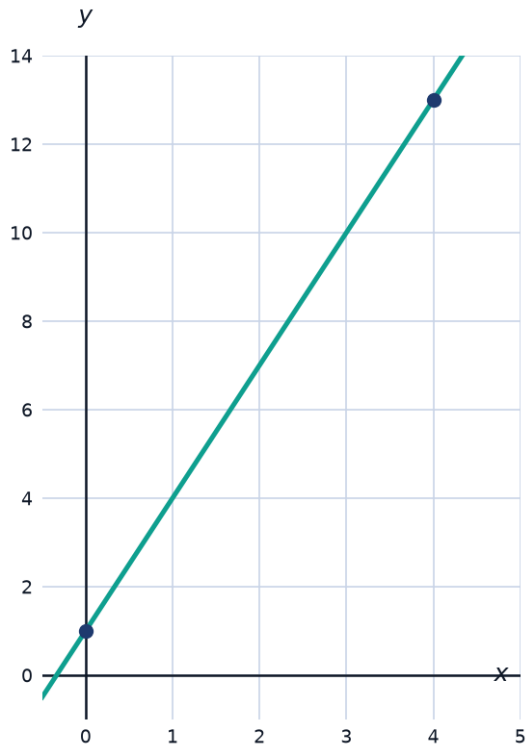
Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$ (improper at the lower limit).



© ClipSAT — clipsat.org | All Rights Reserved

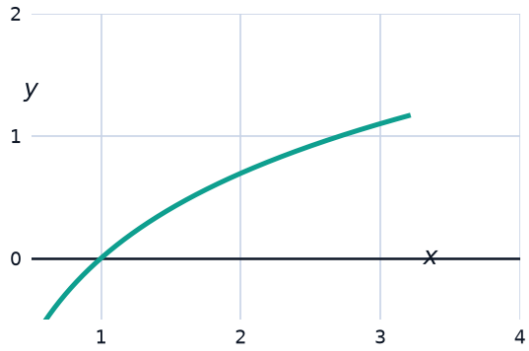
6 [Foundational] • Target: More arc length practice

Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.



7 [Foundational] • Target: More arc length practice

Set up (do not evaluate) the arc length integral for $y = \ln x$ on $[1, e]$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

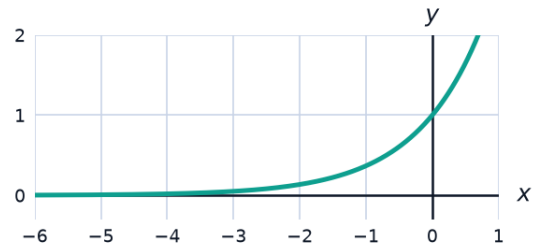
8 [Intermediate] • Target: More improper integrals

Evaluate $\int_1^\infty \frac{1}{x^{1.5}}\,dx$, or state that it diverges.

© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: More improper integrals

Evaluate $\int_{-\infty}^{0} e^x \, dx$, or state that it diverges.



10 [Intermediate] • Target: The p-integral test

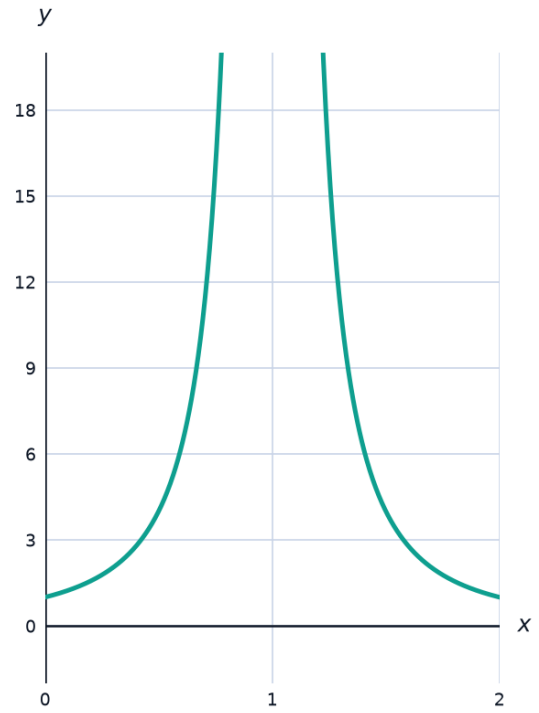
For $\int_{1}^{\infty} \frac{1}{x^p} \, dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).

11 [Intermediate] • Target: The p-integral test

Use the p-integral result to quickly determine whether $\int_{1}^{\infty} \frac{1}{x^{0.9}} \, dx$ converges.

12 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Evaluate $\int_{0}^{2} \frac{1}{(x-1)^2} \, dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).



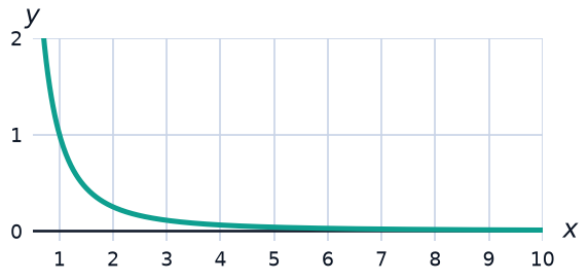
13 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Evaluate $\int_{-1}^{1} \frac{1}{x^{1/3}} \, dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: Visualizing an improper integral as a limiting area

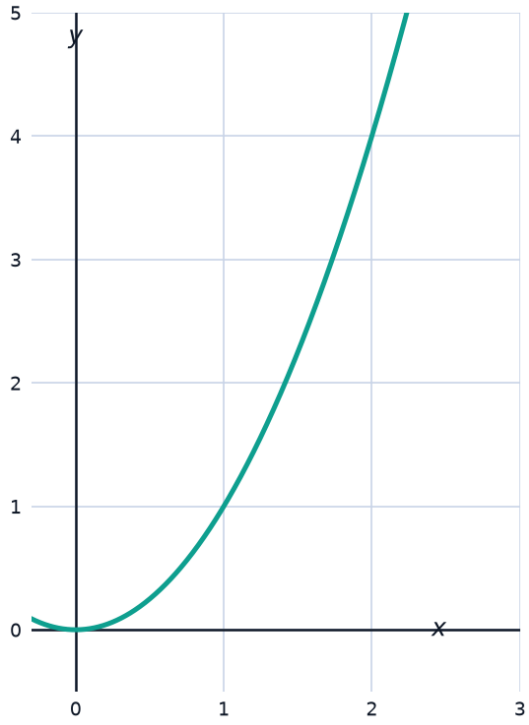
The graph shows $f(x) = \frac{1}{x^2}$ on $[1, 10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^{\infty} \frac{1}{x^2}\,dx = 1$ (found earlier).



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Arc length of a more involved curve • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

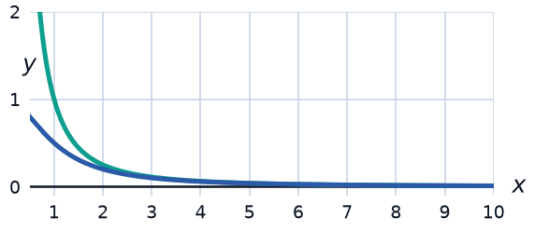
Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.



© ClipSAT — clipsat.org | All Rights Reserved

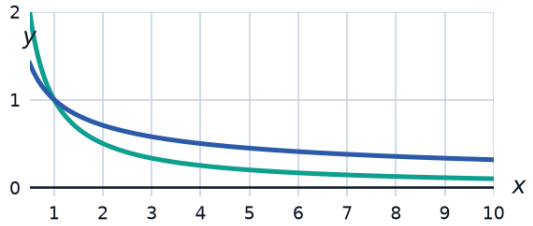
16 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Use a comparison with $\int_1^\infty \frac{1}{x^2}\,dx$ (known to converge) to argue that $\int_1^\infty \frac{1}{x^2+1}\,dx$ also converges, without evaluating it.



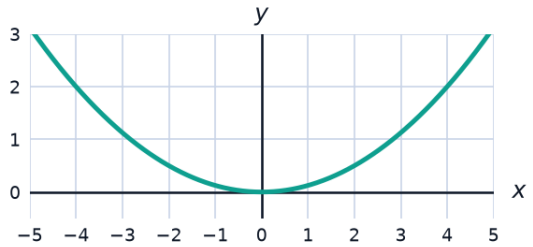
17 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Use a comparison with $\int_1^\infty \frac{1}{x}\,dx$ (known to diverge) to argue that $\int_1^\infty \frac{1}{\sqrt{x}}\,dx$ also diverges.



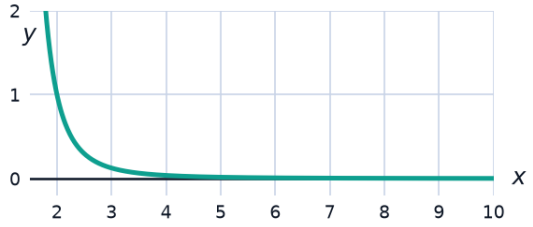
18 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 6.01 Related Rates

A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).



19 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 6.02 Optimization and Curve Sketching

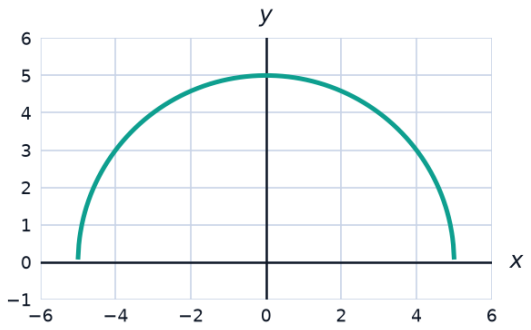
Evaluate $\int_2^\infty \frac{1}{(x-1)^3}\,dx$, or state that it diverges.



© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 7.01 Integration by Substitution

A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x-axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arc Length of a Plane Curve

State the formula for the arc length L of $y = f(x)$ from $x = a$ to $x = b$.

a. $L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$

b. $L = \int_a^b \sqrt{1 + f'(x)}\, dx$

c. $L = \int_a^b [1 + f'(x)]\, dx$

d. $L = \int_a^b \sqrt{[f(x)]^2 + [f'(x)]^2}\, dx$

22 [Foundational] • Target: Arc Length of a Plane Curve

Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

a. $\frac{14}{3}$

b. 7

c. $\frac{16}{3}$

d. 8

23 [Foundational] • Target: Arc Length of a Plane Curve

Find the arc length of the line segment $y = \frac{4}{3}x + 2$ from $x = 0$ to $x = 3$.

a. 5

b. 4

c. 6

d. $\sqrt{7}$

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Improper Integrals of Type I (Infinite Bounds)
Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$.

a. 1

b. ∞ (diverges)

c. 1/2

d. 2

25 [Foundational] • Target: Improper Integrals of Type I (Infinite Bounds)
Evaluate $\int_0^\infty e^{-2x}\,dx$.

a. 1/2

b. 1

c. ∞

d. 2

26 [Foundational] • Target: Improper Integrals of Type I (Infinite Bounds)
For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p}\,dx$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

27 [Foundational] • Target: Improper Integrals of Type II (Discontinuous Integrands)
Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

a. 2

b. 1

c. ∞ (diverges)

d. 1/2

28 [Foundational] • Target: Improper Integrals of Type II (Discontinuous Integrands)
Evaluate $\int_0^2 \frac{1}{x-1}\,dx$.

a. Diverges

b. 0

c. $\ln 2$

d. ∞

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Improper Integrals of Type II (Discontinuous Integrands)

For what values of p does $\int_0^1 \frac{1}{x^p}\,dx$ converge?

a. $p < 1$

b. $p > 1$

c. $p \le 1$

d. $p > 0$

30 [Intermediate] • Target: Comparison Test for Improper Integrals

Determine the convergence of $\int_1^\infty \frac{\sin^2(x)}{x^2}\,dx$.

a. Converges by comparison with $\frac{1}{x^2}$

b. Diverges

c. Oscillates without limit

d. Converges to 0

31 [Intermediate] • Target: Comparison Test for Improper Integrals

Determine the convergence of $\int_1^\infty \frac{1}{\sqrt{x^3+1}}\,dx$.

a. Converges (behaves like $x^{-3/2}$ as $x \to \infty$)

b. Diverges (behaves like $x^{-1/2}$)

c. Diverges to $-\infty$

d. Cannot be tested

32 [Intermediate] • Target: Comparison Test for Improper Integrals

Evaluate $\int_{-\infty}^\infty \frac{1}{1+x^2}\,dx$.

a. π

b. 2π

c. $\pi/2$

d. Diverges

33 [Intermediate] • Target: Improper Integrals of Type II (Discontinuous Integrands)

Comparison test for improper integrals: $0 \le f(x) \le g(x)$, $\int_a^\infty g$ converges implies:

a. $\int_a^\infty f$ converges

b. $\int_a^\infty f$ diverges

c. $f = g$

d. nothing

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Comparison Test for Improper Integrals

Evaluate the improper integral $\int_0^{\infty} \frac{1}{x^2+9}\,dx$.

a. $\frac{\pi}{3}$

b. ∞

c. $\frac{\pi}{6}+1$

d. $\frac{\pi}{6}$

35 [Intermediate] • Target: Arc length

Evaluate the improper integral $\int_0^{\infty} xe^{-3x^2}\,dx$.

a. $\frac{1}{3}$

b. ∞

c. $\frac{1}{6}$

d. $\frac{7}{6}$

36 [Intermediate] • Target: Improper integrals

Improper integral: $\int_1^{\infty} \frac{1}{x^2}\,dx =$

a. 1

b. ∞

c. 1/2

d. 2

37 [Intermediate] • Target: More arc length practice

Improper double integral: $\int_0^{\infty}\int_0^{\infty} e^{-(x+y)}dx\,dy =$

a. 1

b. 0

c. ∞

d. e

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More improper integrals • Cross-Lesson: 6.01 Related Rates

Integral test: $\sum_{1}^{\infty} f(n)$ converges iff $\int_1^{\infty} f(x)dx$ converges, for f positive decreasing. So $\sum 1/n^2$ converges because $\int_1^{\infty} x^{-2}dx =$

a. 1 (finite)

b. ∞

c. 0

d. 1/2

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: The p-integral test • Cross-Lesson: 6.02 Optimization and Curve Sketching

Find the area inside $r = 10\cos(\theta)$ and outside $r = 5$.

a. 25π

b. $25\sqrt{3} + \frac{50\pi}{3}$

c. $\frac{25\pi}{3}$

d. $\frac{25\sqrt{3}}{2} + \frac{25\pi}{3}$

40 [Challenging] • Target: Improper integrals with infinite discontinuities inside the interval • Cross-Lesson: 7.01 Integration by Substitution

Find the area inside $r = 2\cos(\theta)$ and outside $r = 1$.

a. π

b. $\sqrt{3} + \frac{2\pi}{3}$

c. $\frac{\sqrt{3}}{2} + \frac{\pi}{3}$

d. $\frac{\pi}{3}$

41 [Challenging] • Target: Visualizing an improper integral as a limiting area • Cross-Lesson: 7.02 Integration by Parts

Find the area inside $r = 8\cos(\theta)$ and outside $r = 4$.

a. 16π

b. $16\sqrt{3} + \frac{32\pi}{3}$

c. $\frac{16\pi}{3}$

d. $8\sqrt{3} + \frac{16\pi}{3}$

42 [Challenging] • Target: Arc length of a more involved curve • Cross-Lesson: 8.01 Area Between Curves

Find the area inside $r = 4\cos(\theta)$ and outside $r = 2$.

a. 4π

b. $2\sqrt{3} + \frac{4\pi}{3}$

c. $4\sqrt{3} + \frac{8\pi}{3}$

d. $\frac{4\pi}{3}$

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 8.02 Volumes of Revolution

Find the arc length of $y = \cosh(x)$ from $x = 0$ to $x = \ln(3)$.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. $\frac{7}{3}$

d. $\frac{8}{3}$

44 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 9.01 Separable Differential Equations

Find the arc length of $y = \frac{x^3}{6} + \frac{1}{2x}$ from $x = 1$ to $x = 3$.

a. 4

b. $\frac{17}{3}$

c. $\frac{28}{3}$

d. $\frac{14}{3}$

45 [Challenging] • Target: Arc Length of a Plane Curve • Cross-Lesson: 1.01 Precalculus Review for Calculus

The improper integral $\int_0^1 \ln x \, dx$:

a. converges to -1

b. diverges to $-\infty$

c. converges to 0

d. converges to 1

© ClipSAT — clipsat.org | All Rights Reserved